OBJECTIVE–REQUIREMENT–TEST TRACEABILITY

Purpose. This file maps each specific objective from Chapter One to the requirements that operationalise it, the code that implements them, the design elements that explain them, the tests that demonstrate them, and the conclusion that reports the result. It is the audit trail supporting the claim in Chapter Six that no conclusion rests on untested functionality.

References: `Project Proposal- 169613.pdf` §1.3.1 (objectives), Chapter 4 of this documentation set (requirements and design), Chapter 5 (implementation and testing), Chapter 6 (conclusions), repository at `fix/sbms-completion-work` @ `d6b5032`.

Status vocabulary: V = verified by an executed test or recorded observation; **P** = partially verified; **N** = implemented but not tested; **X** = not implemented (by scope or as a recorded gap). Where a requirement is implemented and exercised incidentally but the property it names is never asserted, the row is marked **P** and the missing assertion is named; `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` uses the same verdict for the same requirement, so the two files can be read side by side without reconciling numbers.

---

1. Objective → requirement → evidence map

Objective (i): investigate the existing challenges of manual registration and tracking

Element	Content
Requirements serving the objective	The register as a whole encodes the identified problems; the objective itself is documentary rather than functional
Implementation evidence	Chapter Two document analysis (Auditor-General, Controller of Budget, NG-CDF); requirement register derived from it in Chapter 4 §4.2.1
Design elements	Chapter 4 §4.3.1 (context diagram, which fixes the scope of the problem addressed)
Test evidence	Not applicable — investigative, not behavioural
Conclusion	Chapter 6 §6.2.3: **partially achieved** (secondary evidence only; no fieldwork)
Status	**P**

Objective (ii): design architecture, database schema and ML pipeline

Requirement / design element	Implementation	Demonstration	Status
Three-tier architecture (Ch.4 §4.4.1, Fig. 4.12)	`backend/src/index.js`, `backend/src/routes/*`, `frontend/src/**`, `ml-service/predict.py`	Live deployment checks (Ch.5 §5.3.12); `api.smoke.test.js` 3/3	**V**
Relational schema with integrity constraints (Ch.4 §4.6, Fig. 4.10)	`db/schema.sql`, 14 migration files	`schemaReadiness.test.js` (5); `workflow.test.js` (11) exercise the constraints; recorded 12 tables / 56 constraints	**V**
Six-feature ML pipeline shared by training and serving (Ch.4 §4.3.11, §4.4.4)	`ml-service/feature_engineering.py`, `predict.py`	`ml-service/tests` 33/33 including feature-consistency and artefact loading; `mlPredictor.integration.test.js`	**V**
Random Forest classifier as specified in the proposal	Not in production; research candidate only	`ML_RELIABILITY_REPORT.md` records the candidate's metrics	**X (deviation, Ch.6 §6.2.4)**
Deployment architecture (Ch.4 §4.7, Fig. 4.24)	`Dockerfile`, `.github/workflows/publish-api-image.yml`, Azure resources	Image gates; live deployment checks	**V**

Objective (iii): implement registration, unique identifiers, QR identification and RBAC

Requirement	Implementation	Demonstration	Status
FR-05, FR-06 registration and school constraint	`students.controller.js`, `student.model.js`	`students.test.js` (9); `browser-smoke` checks 2–3	**V**
FR-20 unique QR token and image	`students.controller.js:58-60`, `publicAppUrl.js`	`verification.test.js` checks 1–3	**V**
FR-21 public verification, existence only	`verification.controller.js`	`verification.test.js` (10), `verificationRoute.test.js` (4), `qr-smoke` 9/9	**V**
FR-01 to FR-04 authentication and RBAC	`auth.controller.js`, `middleware/auth.js`, `requireRole.js`	`auth.test.js` (4), `roleAuthorization.test.js` (6), `securityMiddleware.test.js` (9)	**V**
FR-22 to FR-24 certificate issue, verification, `not_current`	`certificates.controller.js`, `bursaryCertificate.model.js`	`bursaryCertificates.test.js` (12), `certificateVerificationRoute.test.js` (5)	**V**
Physical QR scanning by a phone	Implemented artefact; no camera run	—	**N (gap G-11)**

Objective (iv): validate, test and evaluate effectiveness

Requirement / claim	Implementation	Demonstration	Status
FR-07 screening on registration	`students.controller.js:69-73`, `mlDuplicateReview.js`	`duplicateFlags.test.js`, `mlPredictor.integration.test.js`, `performance.test.js`	**V** except at registry scale (gap G-12)
FR-08 no automatic rejection	No delete or status-change path exists	`workflow.test.js` check 1; `browser-smoke` check 5	**V**
FR-09 provenance per case	`duplicate_flags` columns; migration `20261022`	`duplicateFlags.test.js`	**V**
FR-10 masked identifiers, audited reveal	`duplicateFlags.controller.js:37-42, 434-446`	`duplicateFlags.test.js`	**V**
FR-11 decision with reason, terminal case	`duplicateFlags.controller.js:452-503`	`duplicateFlags.test.js`; `workflow.test.js` checks 4–6	**V**
FR-15, FR-16 duplicate gates on approval and payment	`bursaryAwards.controller.js:173-189`; `disbursements.controller.js`	`workflow.test.js` checks 2–4; `disbursements.test.js` check 4	**V**
FR-17, FR-18 clearance evidence and discrepancy	`bursaryRules.js`; `disbursements.controller.js`	`disbursementReconciliation.controller.test.js` (4); `workflow.test.js` checks 8–9	**V**
FR-19 exception derivation	`bursaryRules.js:32-52`	`workflow.test.js`; `bursaryRules.test.js`	**P** (not all eleven reason codes asserted)
FR-25 dashboard, FR-26 report	`adminDashboard.controller.js`, `reports.controller.js`	`roleAuthorization.test.js` (authorisation); arithmetic untested	**P**
FR-27 audit trail	`services/auditLog.js`	`workflow.test.js` check 10 (ten actions)	**V**
FR-12 eligibility scoring	`services/eligibility.js`	Route and role only	**P** (gap G-17)
Detection effectiveness on real records	—	Synthetic holdout only	**X** with existing evidence (gap G-18)

---

2. Non-functional requirement traceability

NFR	Implementation location	Demonstration	Status
NFR-01 screening ≤ 3 s	Batching in `mlDuplicateReview.js`	`performance.test.js` (50 candidates)	**P** (small registry only)

NFR	Implementation location	Demonstration	Status
NFR-02 password hashing	`auth.controller.js:26`	`auth.test.js`	**V**
NFR-03 JWT signing and expiry	`auth.controller.js:55-59`; `jwtSecret.js`	`auth.test.js`; `jwtSecret.validation.node.cjs`	**V**
NFR-04 role not trusted from token	`middleware/auth.js:20-24`	`roleAuthorization.test.js`	**V**
NFR-05 identifier masking	`duplicateFlags.controller.js:37-42`	`duplicateFlags.test.js`	**V**
NFR-06 no beneficiary data in public responses	`verification.controller.js`; `certificates.controller.js:53-89`	`verification.test.js`; `bursaryCertificates.test.js`	**V**
NFR-07 public rate limits, separate budgets	`rateLimitWindow.js`, `verificationRateLimit.js`, `certificateRateLimit.js`	`verificationRoute.test.js`; `certificateVerificationRoute.test.js`	**V** (in-process only)
NFR-08 sign-in rate limit	`loginRateLimit.js`	`loginRateLimit.test.js`	**V**
NFR-09 body size limit	`index.js:64`	`securityMiddleware.test.js`	**V**
NFR-10 no internal detail in errors	`errorHandler.js`	`securityMiddleware.test.js`	**V**
NFR-11 CORS allowlist	`config/cors.js`; `index.js:40-47`	`securityMiddleware.test.js`	**V**
NFR-12 database-enforced integrity	`schema.sql`; migrations	`workflow.test.js`; `bursaryCertificates.test.js`	**V**
NFR-13 registration serialisation	advisory lock in `students.controller.js:68`	Every registration test passes through the lock; no test asserts isolation	**P** (lock present; no concurrency assertion; gap G-12)
NFR-14 usability	`AppLayout.jsx`; screen components	`browser-smoke` (behaviour, not usability)	**P** (renders and behaves; no usability study; gap G-06)
NFR-15 maintainability	shared feature module; ordered migrations; three test frameworks	Builds and suites	**V**
NFR-16 cost envelope	`render.yaml`; deployment log	Deployment record	**P** (configuration recorded; no billed month reconciled)
NFR-17 human oversight	role-gated decision endpoints	`workflow.test.js` check 11	**V**
NFR-18 recovery	`docs/database-recovery-rehearsal-2026-10-31.md`	Rehearsal record	**P** (no schedule; gap G-19)
NFR-19 browser compatibility	SPA over HTTPS	Chrome only	**P** (gap G-16)
NFR-20 predictable degradation	503 on predictor failure	`students.test.js`	**V**
NFR-21 audit completeness	`auditLog.js`	`workflow.test.js` check 10	**P** (attributed in tested flows; no immutability control)

3. Figure and table traceability

Design element	Requirement(s) it explains	Where the behaviour is demonstrated
Fig. 4.1 context diagram	Scope of FR-01 to FR-37 (what is inside the boundary)	Ch.5 §5.6.2 (boundaries)
Fig. 4.2 use case diagram	FR-01 to FR-30	Map in Ch.4 §4.3.2
Fig. 4.3 registration activity	FR-05 to FR-11	`students.test.js`; `workflow.test.js` check 1
Fig. 4.4 review/approval/payment activity	FR-11, FR-15 to FR-19	`workflow.test.js` checks 2–9

Design element	Requirement(s) it explains	Where the behaviour is demonstrated
Fig. 4.5 class diagram	Whole domain	Suite inventory in Ch.5 §5.4.3
Fig. 4.6 registration SSD	FR-05, FR-07, FR-20	`verification.test.js`; `students.test.js`
Fig. 4.7 review sequence	FR-10, FR-11	`duplicateFlags.test.js`
Fig. 4.8 approval-to-certificate sequence	FR-13 to FR-18, FR-22 to FR-24	`workflow.test.js`; `bursaryCertificates.test.js`
Fig. 4.9 public verification sequence	FR-21, FR-23, FR-24	`verification.test.js`; `certificateVerificationRoute.test.js`
Fig. 4.10 ERD	FR-09 to FR-27 (storage)	`schemaReadiness.test.js`
Fig. 4.11 ML pipeline	FR-07, FR-09	`ml-service/tests` (33); `duplicateAggregation.test.js`
Fig. 4.12 architecture	All	Deployment checks
Figs. 4.13–4.23 screens	FR-01 to FR-26 (interface realisation)	`browser-smoke` 23/23; `qr-smoke` 9/9; `registration-fields-smoke` 11/11
Fig. 4.24 deployment	NFR-07, NFR-16	Deployment log §1–§2

4. Test suite → requirement coverage matrix

Cell values give the number of tests in that suite that bear on the requirement group; a blank means the suite is not relevant. Suite totals sum to 151.

Suite (tests)	Auth / RBAC	Registration	Screening	Review	Awards	Payments	QR	Certificates	Reporting	Ops
`auth` (4)	4									
`students` (9)	2	5	2							
`duplicateFlags` (9)	1		1	6		1				
`duplicateAggregation` (13)			13							
`mlDuplicateReview` (8)			8							
`predictorDateNormalization` (6)			6							
`duplicateCheck` (15)		1	14							
`mlPredictor.integration` (1)			1							
`workflow` (11)	1		2	3	2	3				
`bursaryAwards.controller` (5)	1			1	3					
`disbursements` (4)	1			1		2				
`disbursementReconciliation.controller` (4)						4				
`bursaryRules` (4)						4				
`bursaryCertificates` (12)	1				2			9		
`certificateVerificationRoute` (5)								5		
`verification` (10)							10			
`verificationRoute` (4)							4			
`roleAuthorization` (6)	5								1	
`securityMiddleware` (9)	3									6
`schemaReadiness` (5)										5

Suite (tests)	Auth / RBAC	Registration	Screening	Review	Awards	Payments	QR	Certificates	Reporting	Ops
`loginRateLimit` (1)	1									
`performance` (1)			1							
`kenyanSyntheticDataset` (4)			4							
`health` (1)										1

---

5. Conclusion → evidence traceability

Conclusion (Chapter 6)	Evidence base	Gaps
§6.2.1 problem addressed	Ch.1 (as approved), Ch.2 literature, Ch.4 §4.2.1 requirement derivation	—
§6.2.2 purpose and DSR approach	Ch.3 methodology (proposal §3.2–§3.3); artefact and evaluation in Ch.4–5	—
§6.2.3 objective (i) partially achieved	Ch.4 §4.2.1; absence of primary fieldwork	G-06 (no user study)
§6.2.4 objective (ii) achieved with deviation	Ch.4 §4.4–§4.7; `feature_engineering.py`; `predict.py`; migration inventory	G-21
§6.2.5 objective (iii) achieved	Ch.5 §5.3.2–§5.3.9, §5.3.12; suites listed in §5.4.3; deployment checks	G-11 (camera scan)
§6.2.6 objective (iv) partially achieved	Ch.5 §5.4 (all suites), §5.5.2, §5.5.5	G-12, G-18
§6.2.7 engineering quality	Ch.5 §5.4.5 (security, database), §5.5.3, §5.5.4	G-17
§6.2.8 contribution	Ch.5 §5.5.5 (aggregation finding); Ch.4 §4.4.4	G-18
§6.2.9 boundaries	Ch.5 §5.6.2–§5.6.6	G-12, G-13, G-19

---

6. Traceability exceptions to note in the report

1. **Two requirements are demonstrated only by inspection** (FR-26 report content; FR-19 all reason codes). The report says so rather than marking them verified.
2. **Two requirements have no behavioural test** (NFR-13 concurrency; NFR-14 usability). Both are stated as unverified in Chapter Five and carried into the gap register as G-12 and G-06.
3. **One objective's evidence is documentary only** (objective (i)). Chapter Six states this explicitly.
4. **One proposal requirement is not implemented and is reported as a deviation**, not as an omission (Random Forest; G-21).
5. **The production model and the research artefacts are traced separately** so that no metric from `ml-service/experiments/` or `model/research/` is attributed to the deployed system.